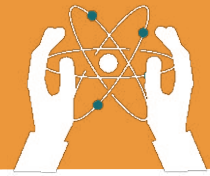


# UNIT 3

## NUTRITION IN ANIMALS

### Learning objectives:

- Students are able to understand:
- Different modes of nutrition.
- Human digestive system with disorders.
- Balanced diet and its importance.

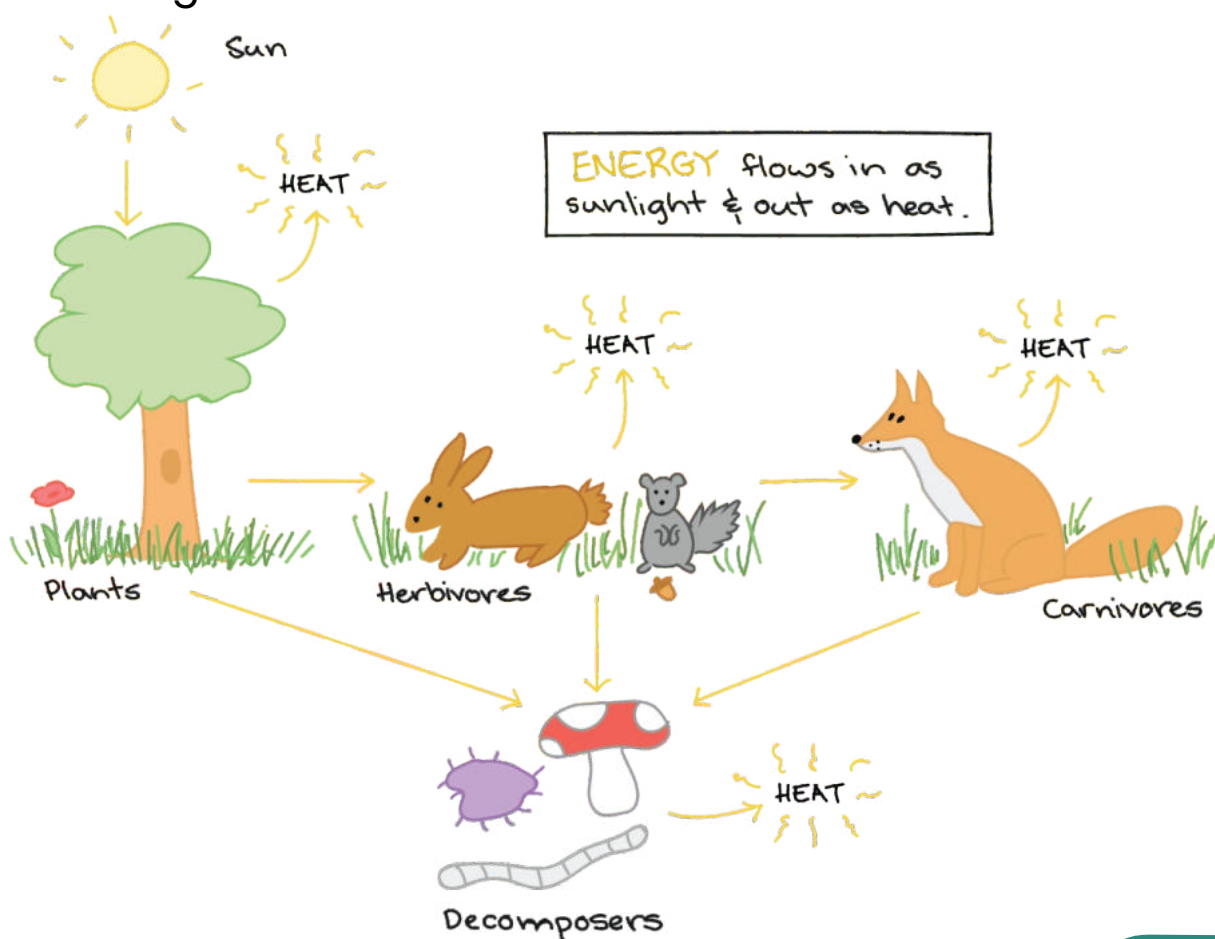


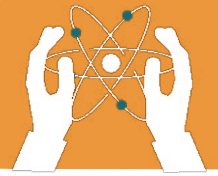
# NUTRITION IN ANIMALS

**Nutrition** is the study of how food and drink affect our bodies with a special regard to the essential nutrients necessary to support human health. It involves understanding the intake of food, how the body utilizes nutrients, and the relationship between diet, health, and disease.

## 3.1 Introduction to Animal Nutrition

Animals rely on food for energy, growth, and repair, as they cannot produce their own food like plants. Nutrition involves ingestion, digestion, absorption, assimilation, and egestion, enabling animals to obtain and use nutrients for survival.





## 3.2 Different Modes of Nutrition in Animals

- **Mainly there are two modes of nutrition in living organisms.**

### 1. Autotrophic Nutrition:

- The type of organisms who can produce their own food.

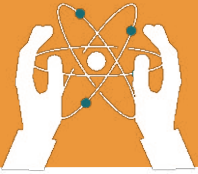
**Example:** Plants using photosynthesis to convert sunlight into energy.

### 2. Heterotrophic Nutrition:

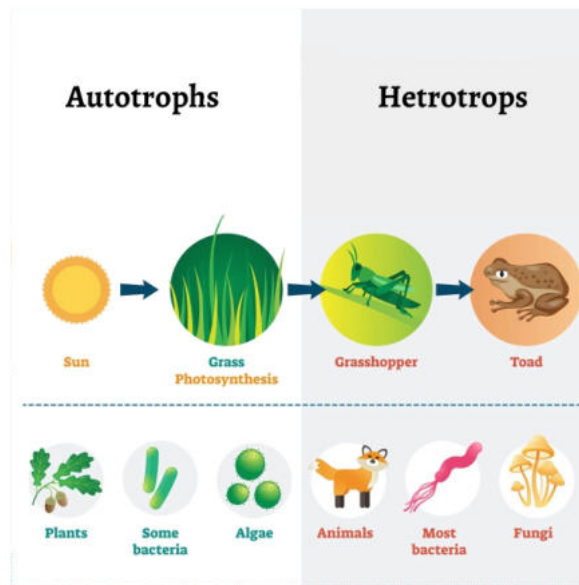
- Type of organisms who depend on other organisms for food.

- **Types:**

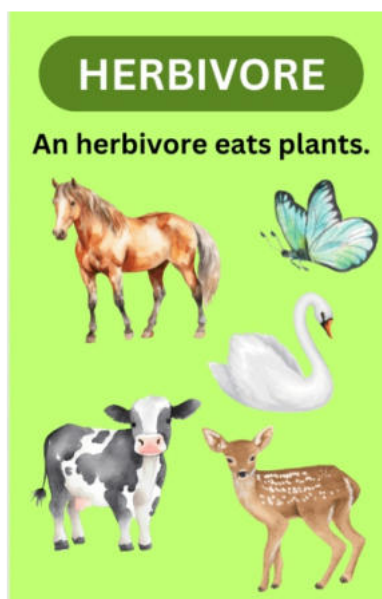
- Such as feeding on dead and decaying matter (e.g., fungi).
- Dependence on a host organism for nourishment (e.g., tapeworms).

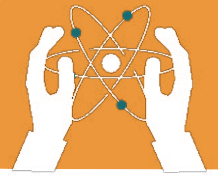


# NUTRITION IN ANIMALS



- **Animals are classified based on their mode of nutrition:**
- **Herbivores:** Animals that eat plants (e.g., cows, deer).
- **Carnivores:** Animals that eat other animals (e.g., wolf, eagles and tigers).
- **Omnivores:** Animals that eat both plants and animals (e.g., humans, bears).





## 3.3 Human Digestive System

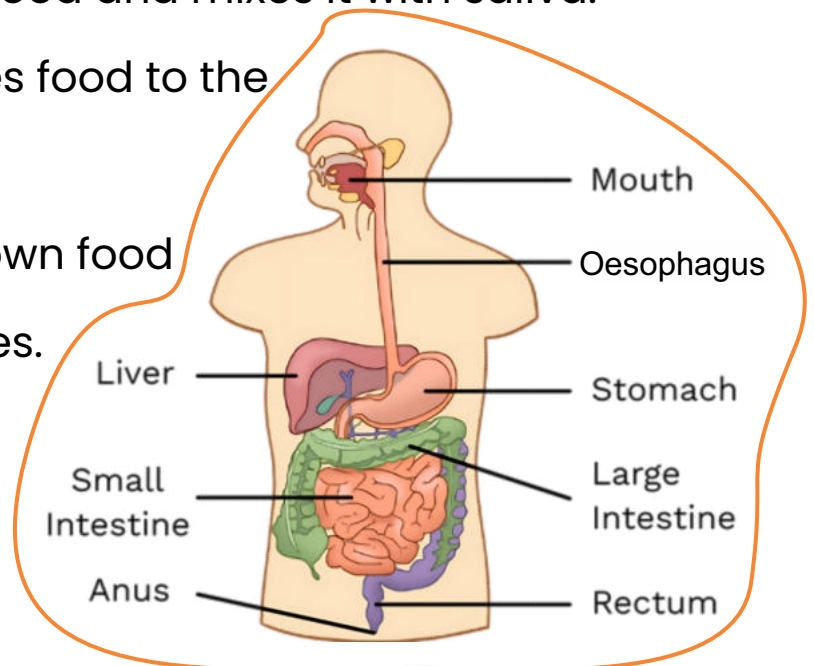
The human digestive system is a complex network of organs that work together to process food. The main components and their functions are as follows:

- **Mouth:** It helps chew food and mixes it with saliva.

- **Oesophagus:** It pushes food to the stomach.

- **Stomach:** It breaks down food with acid and enzymes.

- **Small Intestine:** This absorbs nutrients into the bloodstream.



- **Large Intestine:** This absorbs water and forms waste.

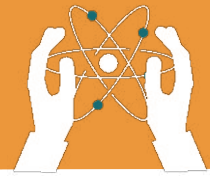
- **Rectum and Anus:** It help in expelling out waste.

Additionally, there are three accessory organs that play crucial roles in the digestive process;

**Liver:** Produces bile for emulsifying fats.

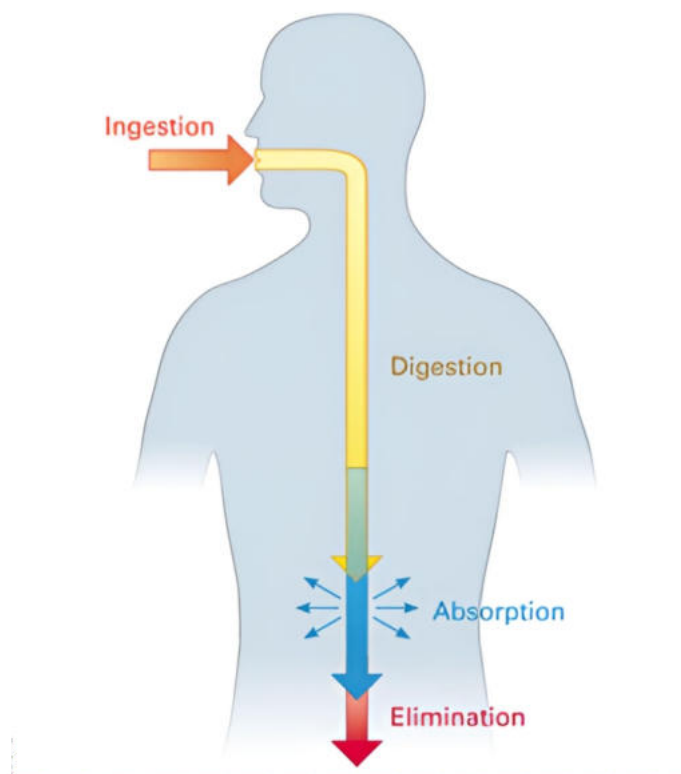
**Gall bladder:** Stores bile from the liver.

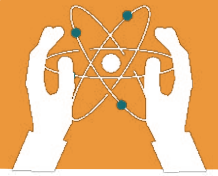
**Pancreas:** Produces pancreatic juice to break down food into smaller molecules.



### 3.4 Process of Digestion in Humans

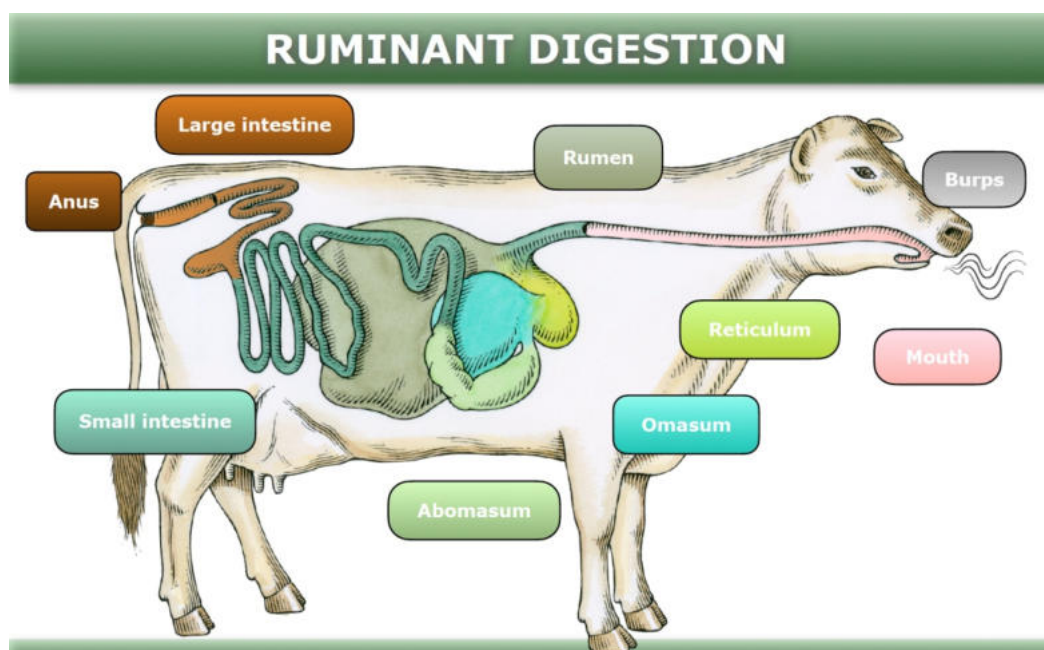
- 1. Ingestion:** Taking in food is termed as ingestion.
- 2. Digestion:** Breaking food into smaller molecules is termed as digestion.
- 3. Absorption:** Nutrients entering the blood is known as absorption.
- 4. Assimilation:** Nutrients used by cells for energy and growth is called as assimilation.
- 5. Egestion:** Removal of undigested food from body is called as egestion.





## 3.5 Digestion in Various Animals

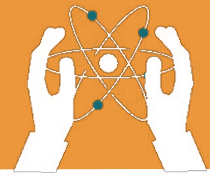
- **Ruminants** (e.g., cows): They have a four-chambered stomach to digest tough plant materials.
- **Amoeba**: These microorganisms use pseudopodia to engulf food and digest it inside food vacuoles.



## 3.6 Balanced Diet and Its Importance

A **balanced diet** provides the necessary nutrients and energy required for optimal health and well-being. It involves consuming the right proportions of various food groups to ensure the body functions effectively.

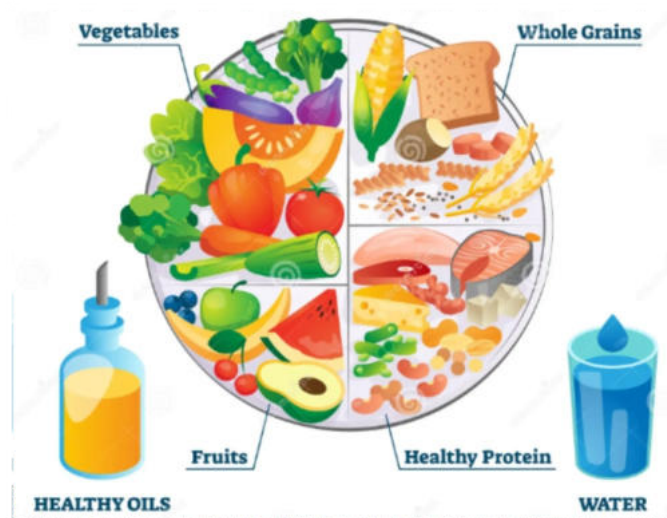




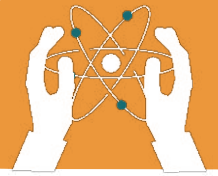
# NUTRITION IN ANIMALS

## Key Components of a Balanced Diet

- 1. Fruits and Vegetables:** They are rich in vitamins, minerals, and fiber.
- 2. Grains:** They include whole grains like brown rice, oats, and whole wheat, which provide essential carbohydrates and fiber.
- 3. Proteins:** This source includes lean meats, poultry, fish, nuts, and seeds, vital for body repair and growth.
- 4. Dairy:** Items like milk, cheese, and yogurt supply calcium and other essential nutrients.
- 5. Fats:** Healthy fats from sources such as avocados, nuts, and olive oil are important, but should be consumed in low quantity.







## 3.7 Common Digestive Disorders

### 1. Indigestion

Indigestion refers to discomfort or pain in the upper abdomen, often occurring during or after eating.

- **Common Causes**

- Consumption of fatty, spicy, or acidic foods; overeating; and excessive caffeine or alcohol intake.

### 2. Constipation

Constipation refers difficult bowel movements, often resulting in hard, dry stools.

- **Common Causes**

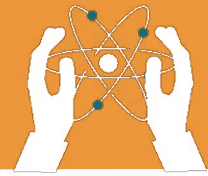
- Low fiber intake and inadequate hydration.

### 3. Diarrhoea

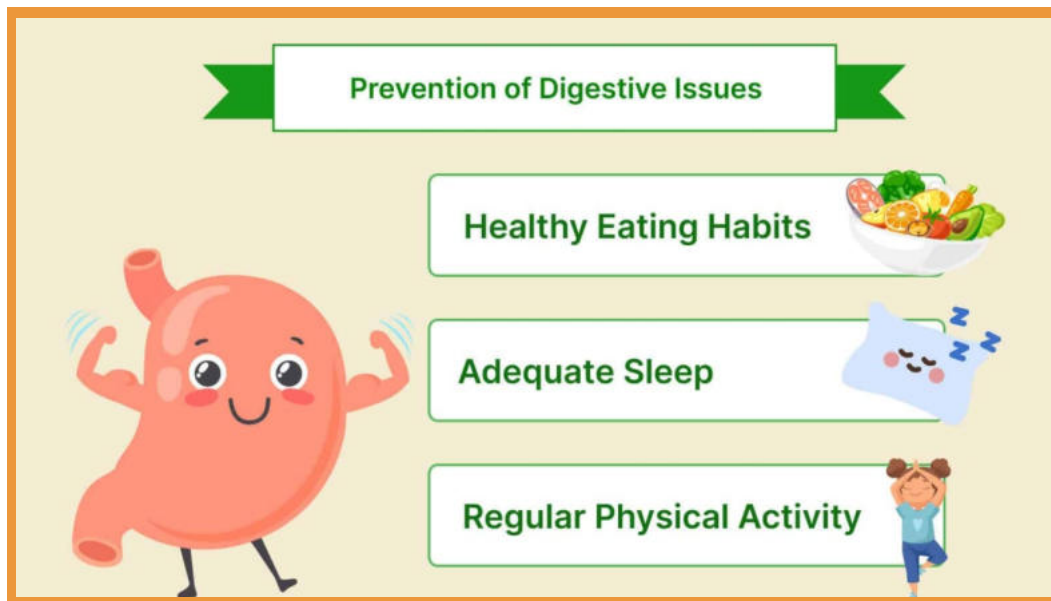
- Diarrhea is characterized by the frequent passage of loose or watery stools from body.

- **Common Cause**

- Viruses such as norovirus and rotavirus are frequent culprits.



## NUTRITION IN ANIMALS

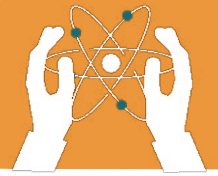


### EXERCISES

#### Fill in the blanks.

1. The process of removing undigested food is called \_\_\_\_\_.
2. Animals that eat both plants and animals are called \_\_\_\_\_.
3. \_\_\_\_\_ is the first stage of digestion.
4. The \_\_\_\_\_ intestine absorbs nutrients from food.
5. A \_\_\_\_\_ diet includes all essential nutrients.
6. \_\_\_\_\_ helps break down food in the stomach.
7. \_\_\_\_\_ are animals with a four-chambered stomach.
8. Amoeba uses \_\_\_\_\_ to capture food.

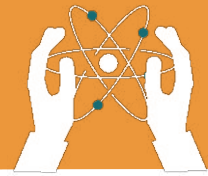
## NUTRITION IN ANIMALS



**Tick(✓) for True and cross (x) for False statements**

1. Herbivores eat both plants and animals.
2. Digestion begins in the stomach.
3. The small intestine is responsible for absorbing nutrients.
4. Carnivores eat only plants.
5. Constipation is caused by too much water in the diet.
6. Assimilation involves using nutrients for energy and growth.
7. Indigestion can result from overeating.
8. Ruminants have a single-chambered stomach.

☐☐☐☐☐☐☐☐



## NUTRITION IN ANIMALS

### Choose the correct answer

**1. The process of taking in food is called:**

- a) Digestion
- b) Ingestion
- c) Assimilation
- d) Egestion

**2. Nutrients are absorbed in the:**

- a) Stomach
- b) Large intestine
- c) Small intestine
- d) Oesophagus

**3. Animals that eat only plants are called:**

- a) Omnivores
- b) Carnivores
- c) Herbivores
- d) None of these

**4. The organ that produces bile is the:**

- a) Stomach
- b) Liver
- c) Pancreas
- d) Small intestine

**5. Which is a digestive disorder?**

- a) Fever
- b) Diarrhoea
- c) Headache
- d) Cough

**6. The first step in digestion is:**

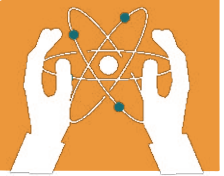
- a) Absorption
- b) Ingestion
- c) Egestion
- d) Assimilation

**7. Amoeba digests food in:**

- a) Mouth
- b) Food vacuole
- c) Stomach
- d) Intestine

**8. A balanced diet includes:**

- a) Only carbohydrates
- b) All nutrients
- c) Only proteins
- d) Only fats



## Answer the Following Questions:

### Short-Answer Questions

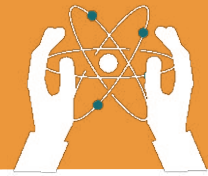
1. What is nutrition and how many modes of nutrition are there in living organism?
2. Write the difference between autotrophic and heterotrophic nutrition.
3. Define a balanced diet and its importance.
4. What is the function of the small intestine?
5. Write the classification of animals based on their modes of nutrition.

### Long-Answer Questions

1. Describe the process of digestion in humans.
2. Describe the components of human digestive system.

### Short Note

1. Write a short note on common digestive disorders.



## NUTRITION IN ANIMALS



### Project-Based Activity



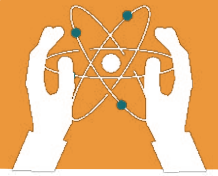
#### ACTIVITY

- Create a **3D model of the human digestive system** using clay or cardboard. Label each organ.
- **Observation Activity:** Chew a piece of bread for 1–2 minutes and note how its taste changes as digestion begins in the mouth. Discuss with the class.



#### Critical Thinking

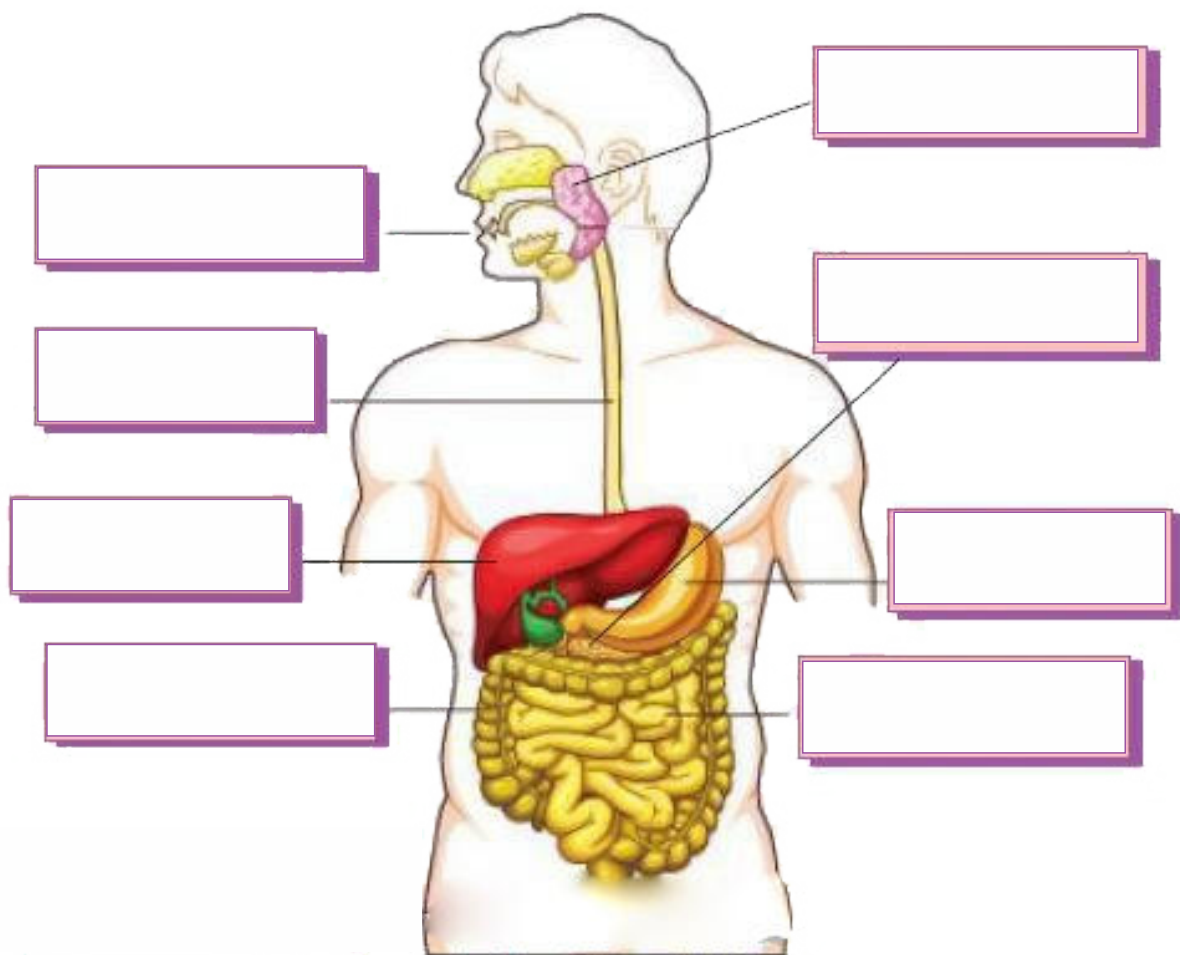
- Why do some animals need more complex digestive systems (e.g., ruminants) compared to others like humans or amoeba?



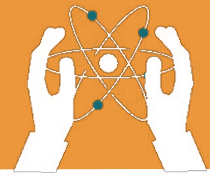
## DIGESTIVE SYSTEM

### 1. Label the digestive system parts.

|                 |                 |       |            |
|-----------------|-----------------|-------|------------|
| Small intestine | Large intestine | Mouth | Oesophagus |
| Stomach         | Salivary gland  | Liver | Pancreas   |







## NUTRITION IN ANIMALS

### 2. Complete the sentences.

- a) The \_\_\_\_\_ is a long tube that pushes the food into the stomach.
- b) The \_\_\_\_\_ is where the digestive process starts.
- c) The \_\_\_\_\_ is like a mixer and it breaks food down into very small pieces.
- d) The \_\_\_\_\_ produces juices that help to transform food into nutrients in the small intestine.
- e) The \_\_\_\_\_ separates the nutrients from the waste.

### WORKSHEET:

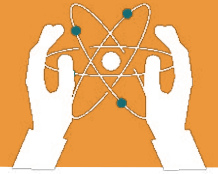
#### Drop into correct place:

### 2. Evaluate the eating habits of the five students based on their meals and answer the nutrition-related questions.

| Amna                            | Hania        | Ali           | Daniyal | Ebbad         |
|---------------------------------|--------------|---------------|---------|---------------|
| Diced chicken                   | Lettuce only | Diced chicken | Carrots | Diced chicken |
| Croutons (toasted bread pieces) | Water        | Corn          | Tomato  | Cheese        |
|                                 |              | Sweet pepper  | Lettuce |               |
|                                 |              | Pasta         | Water   |               |
|                                 |              | Potato        |         |               |
|                                 |              | Milk          |         |               |

- a. Which person may suffer from dehydration? \_\_\_\_\_
- b. Whose diet is most balanced? \_\_\_\_\_
- c. Which person may feel the most tired, unfocused, and hungry by the end of the day? \_\_\_\_\_

## NUTRITION IN ANIMALS



**d.** Which person may feel hyperactive right after lunch?

\_\_\_\_\_

**e.** Which two persons may lose the most weight if they eat like this every day? \_\_\_\_\_ and \_\_\_\_\_

**f.** Whose meal has the most protein? \_\_\_\_\_

**g.** Whose meal has the most carbohydrates? \_\_\_\_\_  
and \_\_\_\_\_